Orbit and Galaxy User-Defined Tools for Reproducible, Auditable Computational Biology

Dannon Baker[1,6], **Scott Cain**[2,6], Marius van den Beek[2,6], John Chilton[2,6], Danielle Callan[2,6], Steven Weaver[3], Nate Coraor[2,6], Kelsey Beavers[4], Michael C. Schatz[1,6], Björn Grüning[5,6,7], Sergei L. Kosakovsky Pond[3], and Anton Nekrutenko[2,6]

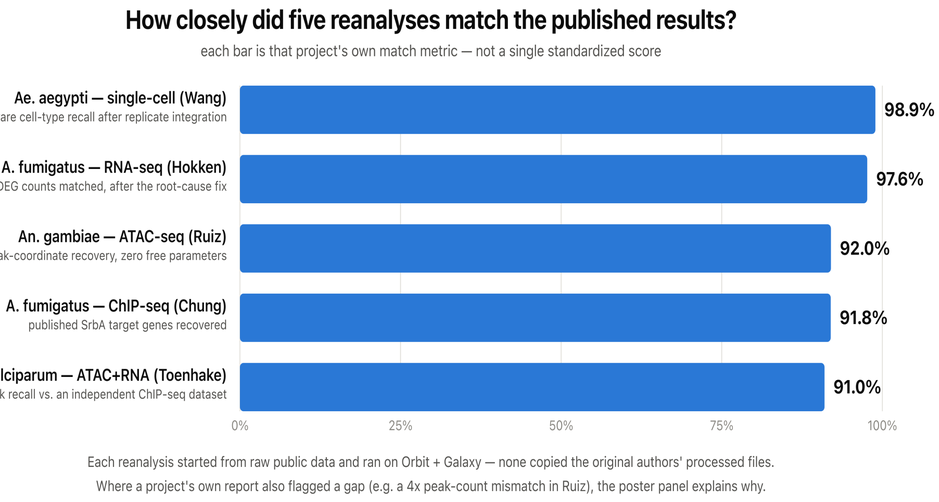
[1]Dept. of Biology, Johns Hopkins University [2]Dept. of Biochemistry & Molecular Biology, Penn State University [3]Institute for Genomics & Evolutionary Medicine, Dept. of Biology, Temple University [4]Texas Advanced Computing Center, UT Austin [5]Dept. of Computer Science, University of Freiburg [6]galaxyproject.org [7]usegalaxy.eu

Five published genomics papers, reanalyzed end-to-end from raw data with an AI agent (Orbit) orchestrating Galaxy and its User-Defined Tools — and an honest account of what matched, what didn't, and why.

Biological Data Science, CSHL, November, 2026

Abstract

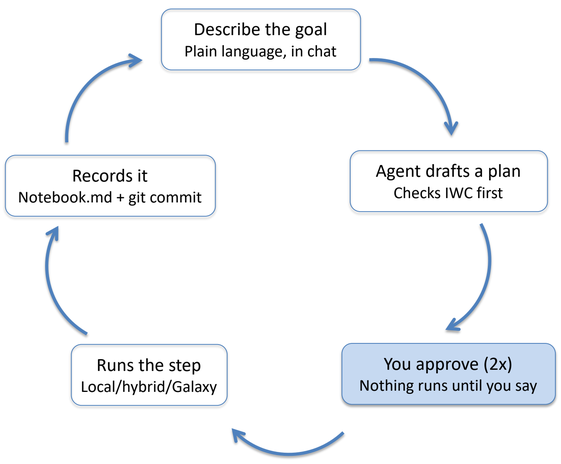
Computational reproducibility in genomics is often undermined not by the standard steps of an analysis — alignment, peak calling, quantification — but by the manuscript-specific "glue" that connects them: custom filtering, integration, and comparison logic that typically lives in untracked local scripts or notebooks tied to a particular machine's software state. We present a workflow that pairs Orbit, an AI-enabled analysis environment, with Galaxy's User-Defined Tools (UDTs) to keep this custom logic inside Galaxy as biocontainer-packaged jobs running on shared HPC or cloud infrastructure, with every input, parameter, and output captured as Galaxy history provenance. Because each UDT is a versioned container rather than a notebook cell tied to a particular interpreter and library snapshot, these steps stay executable and reproducible long after the original analysis, avoiding the software rot that erodes notebook-based workflows as dependencies drift. We apply this workflow to reproduce two published papers directly from raw sequencing data. Reanalyzing Chung et al. (2014), a ChIP-seq/RNA-seq study defining a 97-gene SrbA regulon in Aspergillus fumigatus, Orbit coordinated reference and workflow discovery through BRC-Analytics and ran eight custom UDTs for RNA-seq contrasts, ChIP/RNA integration, and paper-concordance scoring, surfacing an input-control depth imbalance that had distorted peak calling and recovering 89 of 97 published targets outright — with all but one of the remaining eight traced to a decade of annotation coordinate drift rather than lost ChIP-seq signal. Reanalyzing Toenhake et al. (2018), an ATAC-seq/RNA-seq study in Plasmodium falciparum, UDTs handled chromosome renaming in GTF, alignment filtering, and peak-to-gene integration, recovering 94–96% of the paper's published correlated gene-peak pairs against the authors' deposited peak universe. In both cases, routing custom analysis through UDTs — including logic authored and orchestrated by Orbit itself — converted manuscript-specific glue into shareable, containerized Galaxy histories with full job-level provenance. This shows that AI-assisted analysis need not come at the cost of reproducibility or durability: captured as versioned, rerunnable jobs on shared infrastructure rather than as opaque local scripts or notebooks, AI-agent-driven analysis can be at least as inspectable, reproducible, and long-lived as one written entirely by hand.



How a session runs



Orbit, in an actual session — files, chat/plan, and notebook panes



Nothing executes, and nothing is written to the record, until the user approves it — twice.

Tool Source for srba-chip-rna-integrator

```
class: GalaxyUserTool
id: srba-chip-rna-integrator
name: SrbA ChIP RNA integrator
version: 0.1.0
description: Assign Af293 SrbA ChIP peaks to genes and merge 120-minute RNA-seq
  fold-change contrasts.
container: python:3.11-slim
shell_command: |-
  python <<'PY'
  import csv, math, re
  from collections import defaultdict
  from pathlib import Path

  ANNOTATION = Path('${inputs.annotation.path}')
  PEAKS = Path('${inputs.peaks.path}')
  CONTRAST_A = Path('${inputs.dsrba120_all.path}')
  CONTRAST_B = Path('${inputs.dsrb120_all.path}')

  PROMOTER_WINDOWS = {'promoter_1kb': (1000, 200), 'promoter_2kb': (2000, 200)}
  NEAR_TSS_CUTOFF = 2000
  KEY_TARGET_SYMBOLS = {'srbA','srbB','erg3','erg24','erg25','erg25B','cyp51A','erg11A','cyp51B','erg11B','hem13','niiA','alcC','sit1','hyd1'}
  KEY_TARGET_IDS = {'AFUA_2G01260','AFUA_4G03460','AFUA_4G06890','AFUA_7G03740','AFUA_4G04820','AFUA_8G02440','AFUA_1G07480','AFUA_1G12840','AFU
  def parse_gtf_attrs(s):
      d = {}
      for part in s.split(';'):
          part = part.strip()
          if not part or ' ' not in part:
              continue
          k, v = part.split(' ', 1)
          d[k] = v.strip().strip('"')
      return d

  def gene_sort_key(gid):
      m = re.match(r'AFUA_(\d+)G(\d+)', gid or '')
```

Tool Source for srba-chip-rna-integrator

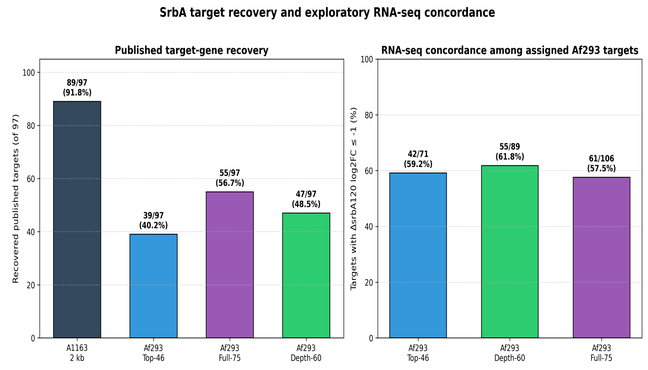
```
outputs:
  - name: top46_overlap
    label: Key target ChIP RNA overlap top46
    type: data
    format: tabular
    from_work_dir: key_target_chip_rna_overlap_top46.tsv
  - name: summary
    label: SrbA ChIP RNA integration summary
    type: data
    format: tabular
    from_work_dir: chip_rna_integration_summary.tsv
  - name: report
    label: SrbA ChIP RNA integration report
    type: data
    format: txt
    from_work_dir: chip_rna_integration_report.txt
help:
  format: markdown
  content: >-
    Assigns Af293 SrbA ChIP-seq MACS2 peaks to nearby genes and joins the
    assignments to 120-minute matched RNA-seq fold-change tables.


    Inputs are an Af293 GTF annotation, a matched-control Af293 MACS2 narrowPeak
    BED, and the Step 9 all-gene tables for WT120 vs dSrbA120 and WT120 vs
    dSrbB120. The tool uses gene-body overlap, promoter windows of 1 kb or 2 kb
    upstream plus 200 bp downstream, and nearest TSS within 2 kb.


    Outputs include all peak-gene assignments, one best peak assignment per
    target gene, targets down in each mutant, canonical/key target overlaps, a
    conservative top-46 peak subset by MACS2 q-value, and a summary/report. Fold
    changes are exploratory n=1 values from the upstream Step 9 tables.
```

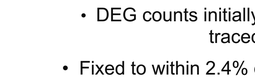
An example UDT that assigns SrbA ChIP-seq peaks to genes and merges them with matched RNA-seq fold-change data in a single run, replacing what would otherwise be a multi-step manual join.

Aspergillus fumigatus · ChIP-seq — Chung et al. 2014



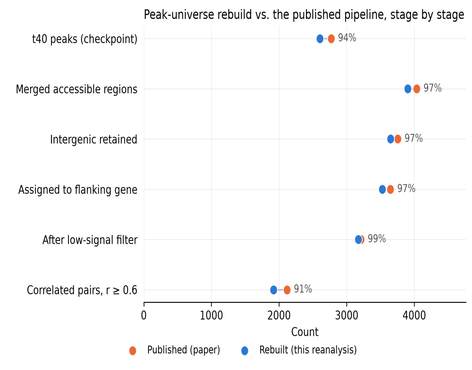
Recovery of the paper's 97 published SrbA target genes (left) and RNA-seq concordance among Af293-assigned targets (right), across four peak-calling designs.

- 8 custom Galaxy UDTs ran the manuscript-specific steps: RNA-seq contrasts, ChIP/RNA integration, and paper-concordance scoring.
- A depth imbalance between input controls — easy to miss in a notebook — surfaced as an inspectable Galaxy job and was corrected.
- **89 of 97 published SrbA target genes recovered outright; all but one of the rest traced to a decade of annotation coordinate drift, not lost signal.**



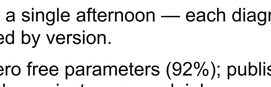
Caught its own mistake: a "verified" tool link meant nothing — re-audited by exporting the real UDT definitions.

Plasmodium falciparum · ATAC-seq + RNA-seq — Toenhake et al. 2018



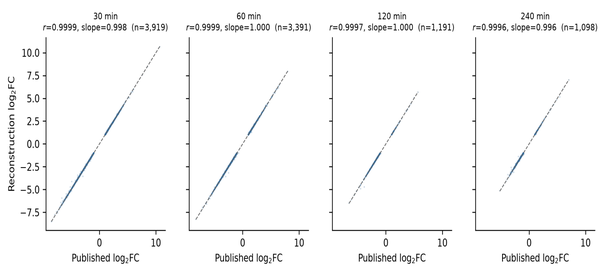
Peak-universe rebuild vs. the published pipeline, stage by stage — published counts (orange) vs. this reanalysis (blue), with percent match labeled at each step.

- UDTs handled chromosome renaming, alignment filtering, and peak-to-gene integration — across two genome annotation releases, run side by side on purpose.
- **94–96% of the paper's correlated gene–peak pairs recovered using the authors' deposited peaks; 91% recall / 96% precision rebuilding the peak universe from raw reads alone.**
- An apparent 4× shortfall turned out to be a smoke-test tool mistaken for a peak caller — not a flaw in the published method.



Declared the paper "irreproducible," then found its own placeholder peak-caller — a named, versioned UDT — and reversed the verdict in 24 hours.

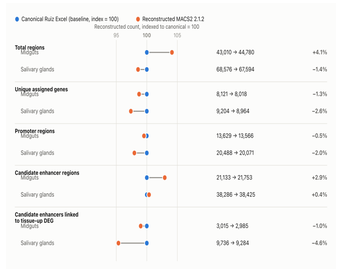
A. fumigatus RNA-seq — Hokken 2018



Reanalysis log2FC vs. the paper's log2FC at each timepoint (r, slope shown); dashed line is y = x.

- DEG counts initially missed the paper by 31–56% at later timepoints — traced to a modeling choice, not bad data.
- Fixed to within 2.4% of published counts; AspGD's retirement replaced with custom GO-enrichment UDTs.
- Checked its own workflow design against the IWC registry — and overturned its own earlier assumption when the evidence didn't support it.

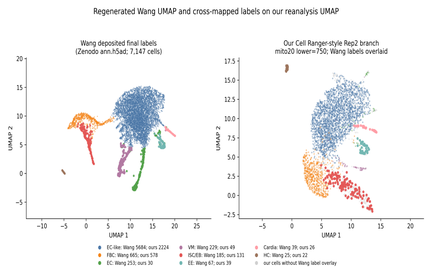
An. gambiae ATAC-seq — Ruiz 2021



Canonical (paper) vs. independently reconstructed region counts, across five regulatory summaries.

- One UDT failed three different ways in a single afternoon — each diagnosed and fixed, logged by version.
- Peak coordinates reproduced with zero free parameters (92%); published peak counts did not — and the project says so plainly.
- Pre-registered a blind cross-tissue validation test rather than trust a fitted parameter — and reported "inconclusive" when that was the honest answer.

Ae. aegypti single-cell — Wang 2024



Left: UMAP from Wang et al.'s deposited labels. Right: this reanalysis's UMAP with those labels cross-mapped onto matching barcodes.

- Reanalysis filed 3 real upstream Galaxy / IWC bugs and contributed 3 reusable Skills — benefits beyond this one paper.
- A background process once silently erased notebook entries — recovered from the raw activity log two months later.
- Built a Cell Ranger-style reference for a mosquito genome Galaxy had no recipe for — down to fixing non-standard mitochondrial gene naming.

5 published papers reanalyzed — 5 organisms, 4 assay types

67+ custom Galaxy tools built where none existed

3 real Galaxy / IWC bugs found and fixed as a side effect

48+ dated git commits forming an auditable project record

AI-assisted analysis doesn't have to trade away reproducibility.

Captured as versioned, auditable Galaxy jobs rather than opaque local scripts, it can be at least as inspectable and durable as analysis written entirely by hand.

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